REALTIME STREET LIGHTING SYSTEM ACTIVATED BY VEHICLE MOTION USING 8051 MICROCONTROLLER

A MINIPROJECT REPORT

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BONAFIDE CERTIFICATE

Certified that this report titled **“REALTIME STREET LIGHTING SYSTEM ACTIVATED BY VEHICLE MOTION USING 8051 MICROCONTROLLER”** is the bonafide work done by **PREETHI.S (421122104070)** & **VISHNU PRIYA.B (421122104110)** who carried out the work under my supervision. Certified further that to the best of my knowledge the work reported herein does not form part of any other thesis or dissertation on the basis of which a degree or award was conferred on anearlier occasion on this or any other candidate.

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CERTIFICATE OF EVALUATION

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The report for the Mini project-II submitted for the fulfillment of the award of the degree of Bachelor of Electronics and Communication Engineering of IFET College of Engineering (Autonomous), permanently affiliated to Anna University was evaluated and confirmed to be the work done by the above student.

ABSTRACT

The "Realtime Street Lighting System Activated by Vehicle Motion Using 8051 Microcontroller" project presents a smart and energy-efficient approach to street lighting. Traditional street lighting systems consume a significant amount of energy by remaining illuminated throughout the night, regardless of traffic density. This project addresses the issue by implementing a motion-activated lighting system that turns on street lights only when vehicles are detected. The system consists of an 8051 microcontroller, an array of infrared (IR) sensors, a relay module, and LED lights representing street lamps. Each IR sensor is positioned along the road to monitor specific sections for vehicle movement. When a vehicle enters a section, the corresponding IR sensor detects the motion and sends a signal to the 8051 microcontroller. In response, the microcontroller activates the relay module to turn on the associated street light. Once the vehicle moves past the detection range, the light automatically turns off after a short delay, conserving energy. This project demonstrates an intelligent street lighting system that provides illumination only when necessary, significantly reducing energy consumption and operational costs. By incorporating motion-based activation, the system extends the lifespan of street lights and contributes to environmental sustainability by reducing light pollution and energy waste. This approach is ideal for low-traffic areas, rural roads, and pathways, offering a cost-effective and scalable solution for modern urban infrastructure.

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LIST OF ABREVIATIONS

IR	Infrared
LED	Light Emitting Diode
IOT	Internet of Things
ROM	Read-Only Memory
RAM	Random Access Memory
UART	Universal Asynchronous Receiver-Transmitter
LIDAR	Light Detection and Ranging
APWIMOB	Asia Pacific Conference on Wireless and Mobile
ICITR	International Conference on Information Technology Research
ICRITO	International Conference on Reliability, Infocom Technologies and Optimization
ICCCT	International Conference on Computing and Communication Technologies

CHAPTER 1
INTRODUCTION

1.1 INTRODUCTION

Street lighting is an indispensable part of urban and rural infrastructure, providing safety, security, and convenience during nighttime. Traditional street lighting systems, however, have inherent limitations, as they typically remain illuminated throughout the night, irrespective of the actual need. This results in significant energy wastage, especially in areas with low traffic density or during late-night hours. With the global emphasis on energy conservation and environmental sustainability, there is a pressing need to modernize street lighting systems to make them more efficient and responsive to real-world conditions. The project titled "Realtime Street Lighting System Activated by Vehicle Motion Using 8051 Microcontroller" proposes a smart solution to address these challenges by introducing an automated system that activates street lights only when vehicular motion is detected. The proposed system uses an 8051 microcontroller as its central control unit, working in conjunction with infrared (IR) sensors and a relay module to manage the activation and deactivation of street lights in real time. IR sensors are strategically positioned along the road to monitor specific segments for vehicle movement. When a vehicle enters the detection range of an IR sensor, the sensor generates a signal that is sent to the 8051 microcontroller. The microcontroller processes this input and activates the relay module, which, in turn, powers on the corresponding street light. Once the vehicle exits the detection range, the system automatically turns off the light after a short delay, ensuring that energy is conserved when the road is not in use. This motion-based system eliminates the need for street lights to remain on continuously, reducing electricity consumption and operational costs. One of the key highlights of this project is its energy efficiency. By operating street lights only when required, the system significantly minimizes power consumption, making it a cost-effective alternative to traditional street lighting systems. Additionally, the controlled operation of street lights reduces their overall runtime, thereby extending their lifespan and lowering maintenance and replacement costs. This system also contributes to environmental sustainability by reducing energy demand and the associated carbon footprint. Unlike conventional systems that contribute to light pollution by illuminating areas unnecessarily, this system ensures that lights are activated only when needed, further promoting eco-friendliness. The modular design of this system makes it scalable and adaptable for various real-world applications. It can be implemented in urban streets, highways, rural roads, parking lots, and even private properties. The use of IR sensors ensures reliable detection of vehicles in diverse conditions, while the 8051 microcontroller serves as a robust and efficient processing unit capable of managing multiple inputs and outputs. The relay module facilitates safe and reliable switching of high-powered street lights, ensuring the system's safety and functionality. Moreover, this project aligns with the global vision of smart cities, where technology is leveraged to optimize resources and improve the quality of life. The incorporation of real-time motion detection and automation makes this system a step forward in modernizing urban infrastructure. By integrating cost-effectiveness, energy conservation, and environmental sustainability, the system addresses the need for smarter and more sustainable urban development. In conclusion, the "Realtime Street Lighting System Activated by Vehicle Motion Using 8051 Microcontroller" project offers a practical and innovative approach to revolutionizing traditional street lighting systems. It provides an efficient solution to reduce energy wastage, lower operational costs, and minimize environmental impact while maintaining safety and functionality. This project demonstrates how simple yet effective technologies can be used to achieve significant advancements in infrastructure and sustainability. By addressing key challenges associated with traditional street lighting, this system not only promotes energy efficiency but also sets a benchmark for the development of intelligent solutions in the future.

1.2 OBJECTIVE

The primary objective of the Automatic Street light system is

- **To ensure** that street lights operate only when required, thereby minimizing unnecessary power consumption during low-traffic hours.
- **To automate** the process of activating and deactivating street lights based on real-time vehicle movement.
- **To reduce** operational and maintenance costs by prolonging the lifespan of street lights through controlled usage.
- **To lower** the carbon footprint by conserving energy and minimizing light pollution in both urban and rural areas.
- **To design** a scalable and practical system that can be implemented in various real-world scenarios, such as streets, highways, and parking areas.

1.3ORGANIZATION OF THE PROJECT REPORT

CHAPTER 1: Gives an introduction and describes the objective of the project.

CHAPTER 2: Explains the proposed design of the project with block diagram, circuit and the model.

CHAPTER 3: Give the detailed description of the hardware components used in this project.

CHAPTER 4: Explains the application of this project.

CHAPTER 5: Contains the conclusion.

CHAPTER 2

PROPOSED DESIGN

2.1 BLOCK DIAGRAM

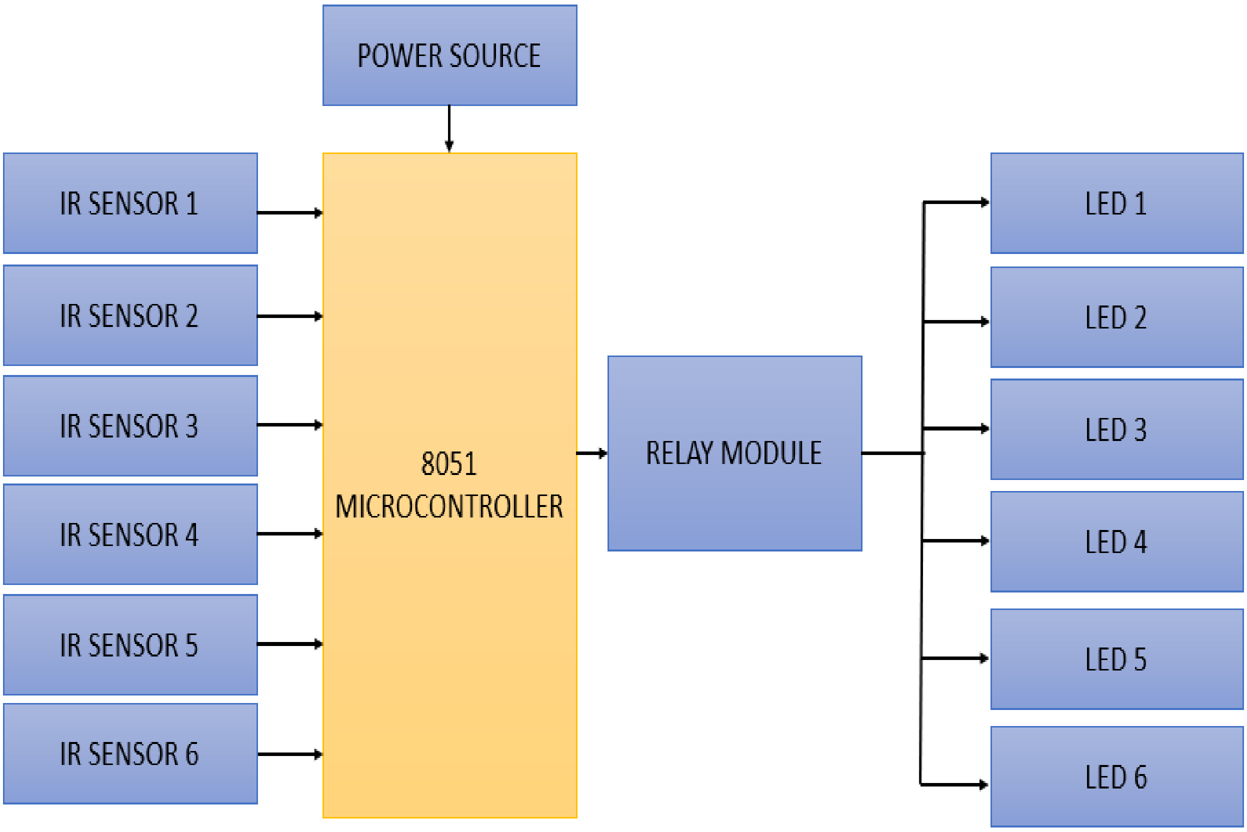

Fig 2.1 Block diagram

The block diagram illustrates the operation of the "Realtime Street Lighting System Activated by Vehicle Motion Using 8051 Microcontroller", which automates street lighting based on vehicle movement. A power source supplies energy to the 8051 microcontroller, IR sensors, relay module, and LEDs. IR sensors detect vehicles and send signals to the microcontroller, which activates the corresponding LED lights via the relay module. Lights are deactivated after a short delay when no vehicles are detected, conserving energy. This efficient design reduces power consumption, operational costs, and environmental impact, while being scalable for applications like highways, streets, and parking lots.

2.2 CIRCUIT DIAGRAM

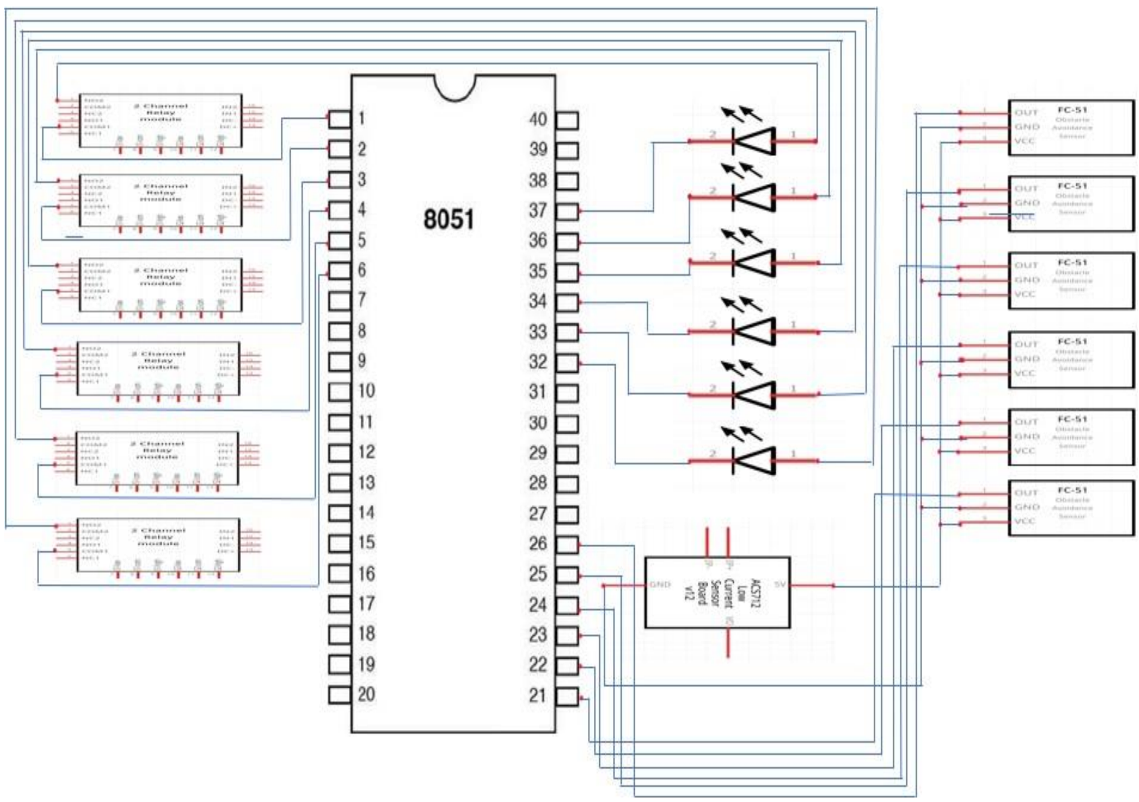

Fig 2.2 Circuit diagram

The circuit is a motion-controlled street lighting system that uses the 8051 microcontroller, IR obstacle avoidance sensors, relays, and LEDs. The 8051 acts as the central controller, processing signals from six IR sensors to detect motion, such as vehicles or pedestrians. Each IR sensor's output pin is connected to specific input pins of the 8051 (e.g., P0.0 to P0.5). Based on the sensor input, the microcontroller activates the corresponding relay by sending a HIGH signal to its control pin via output ports (e.g., P2.0 to P2.5). The relay, powered by a 12V supply, completes the circuit for its connected LED, causing it to turn on. Each relay module ensures isolated switching, while the LEDs serve as streetlights. The 5V power supply powers the 8051, relays, and sensors. When motion is no longer detected, the microcontroller deactivates the relay, turning the LED off to save energy. The system is simple, efficient, and scalable, providing an energy-efficient solution for smart street lighting applications.

2.3 PROPOSED MODEL

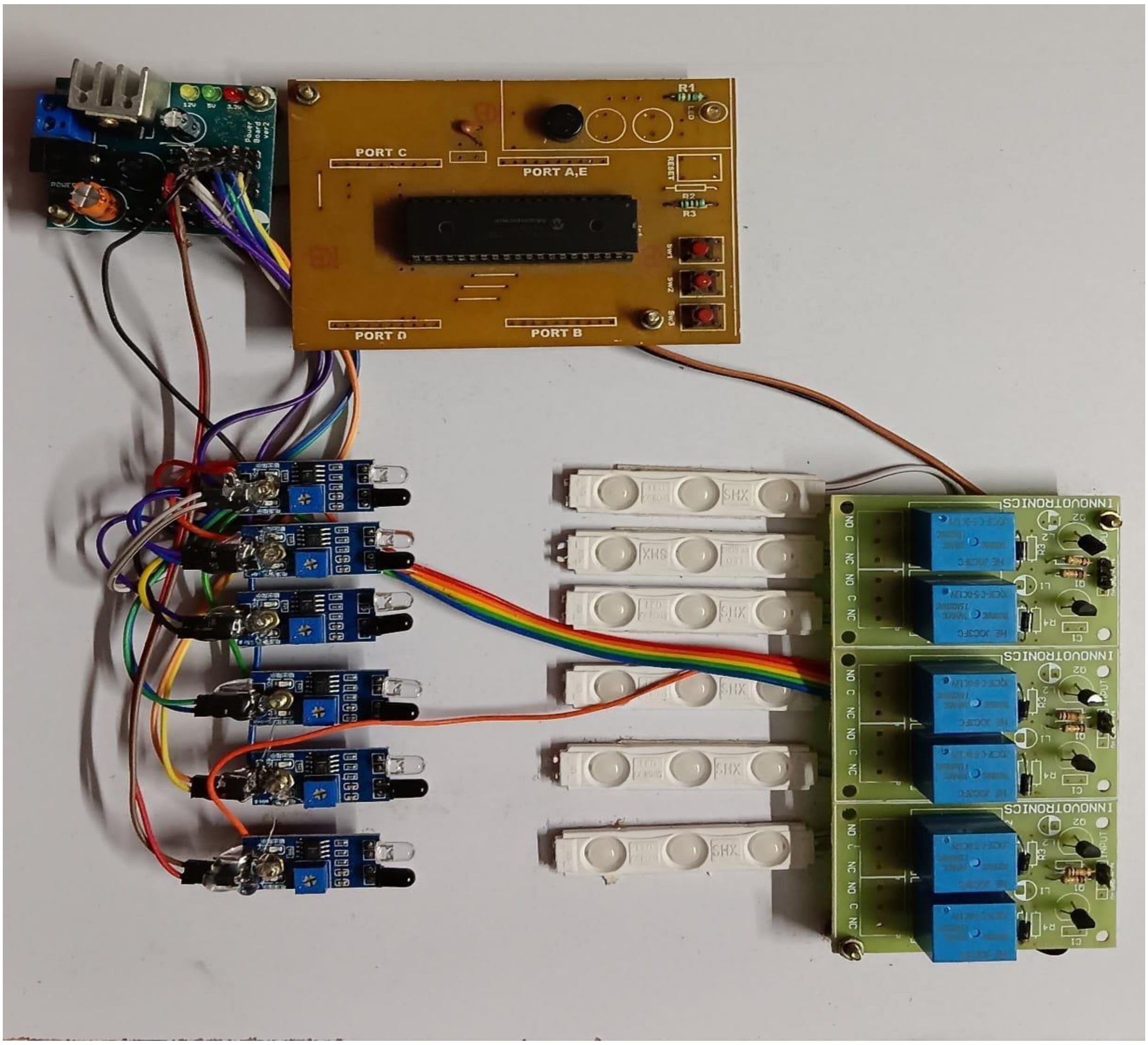

Fig 2.3 Proposed Model

This project, "Realtime Street Lighting System Activated by Vehicle Motion Using 8051 Microcontroller", automates street lighting to enhance energy efficiency. The system uses IR sensors to detect vehicle movement and sends signals to the 8051 microcontroller, which processes the data and activates corresponding LEDs via a relay module. The LEDs light up only when a vehicle is detected, and turn off after a short delay when the vehicle passes, conserving power. A power supply module ensures the proper functioning of all components. This setup is scalable for applications like streets, highways, and parking lots, promoting sustainability and cost efficiency.

2.4 SOURCE CODE

```
#include <reg51.h>
// Define ports for IR sensors and LEDs
sbit IR1 = P1^0;   // IR Sensor 1
sbit IR2 = P1^1;   // IR Sensor 2
sbit IR3 = P1^2;   // IR Sensor 3
sbit IR4 = P1^3;   // IR Sensor 4
sbit IR5 = P1^4;   // IR Sensor 5
sbit IR6 = P1^5;   // IR Sensor 6
sbit LED1 = P2^0;  // LED 1
sbit LED2 = P2^1;  // LED 2
sbit LED3 = P2^2;  // LED 3
sbit LED4 = P2^3;  // LED 4
sbit LED5 = P2^4;  // LED 5
sbit LED6 = P2^5; // LED 6
void delay(unsigned int time)
{
   unsigned int i, j;
   for (i = 0; i < time; i++)
      for (j = 0; j < 1275; j++);
}
void main()
{
   // Initialize all LEDs to OFF
   LED1 = 0;
   LED2 = 0;
   LED3 = 0;
   LED4 = 0;
   LED5 = 0;
   LED6 = 0;
   while (1)
{
      // Check for IR Sensor inputs and turn on corresponding LEDs
      if (IR1 == 1) {
         LED1 = 1; // Turn on LED1
         delay(100); // Wait before turning off
         LED1 = 0;  // Turn off LED1
      }
      if (IR2 == 1) {
         LED2 = 1;
         delay(100);
         LED2 = 0;
```

```
        }
        if (IR3 == 1) {
            LED3 = 1;
            delay(100);
            LED3 = 0;
        }
        if (IR4 == 1) {
            LED4 = 1;
            delay(100);
            LED4 = 0;
        }
        if (IR5 == 1) {
            LED5 = 1;
            delay(100);
            LED5 = 0;
        }
        if (IR6 == 1) {
            LED6 = 1;
            delay(100);
            LED6 = 0;
        }
    }
}
```

CHAPTER 3
HARDWARE AND COMPONENTS

3.1 8051 MICROCONTROLLER

Fig 3.1 8051 Microcontroller

The 8051 microcontroller is a versatile and widely used 8-bit microcontroller initially developed by Intel, known for its simplicity and reliability in embedded systems. It features an 8-bit CPU, 4KB of onboard ROM for program storage, and 128 bytes of RAM for data handling. The 8051 comes equipped with 4 I/O ports, allowing easy interfacing with external devices, as well as two 16-bit timers/counters for timing and event management. It also supports serial communication via UART and has 5 interrupt sources to efficiently handle multiple tasks. Due to its robust design and flexibility, the 8051 microcontroller is commonly used in control and automation applications, such as your real-time street lighting system project.

3.2 IR OBSTACLE AVOIDANCE SENSOR

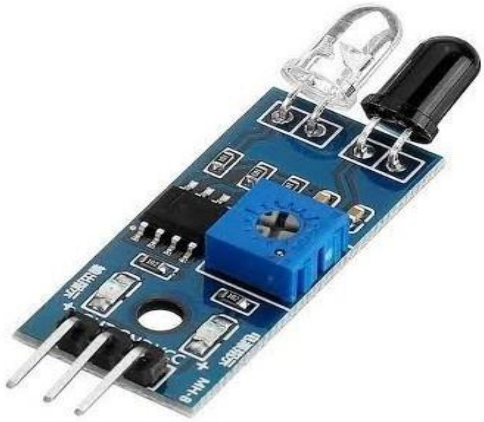

Fig 3.2 IR Sensor

The IR obstacle avoidance sensor is a type of infrared sensor specifically designed to detect obstacles in its path. It uses an infrared emitter (IR LED) and a receiver (photodiode or phototransistor) to identify objects. The sensor works based on the principle of reflection: the IR emitter sends out infrared light, and if an obstacle is present, this light is reflected back toward the receiver. The sensor compares the received signal's intensity and generates an electrical output, which can be processed by a microcontroller to determine the presence and proximity of an obstacle. These sensors typically have an adjustable range, allowing customization based on the application's requirements.

3.3 RELAY MODULE

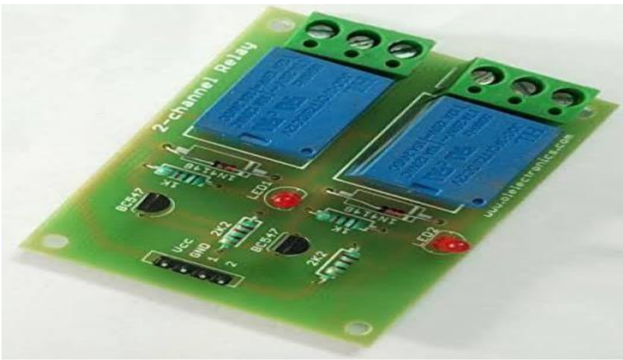

Fig 3.3 Relay Module

A relay module is an electronic device used to control high-voltage or high-current loads with low-voltage signals from a microcontroller or sensor.

It acts as an electrically operated switch, allowing a small electrical signal to control a larger electrical circuit. The module typically consists of a relay (an electromechanical switch), an optocoupler for isolation, and supporting components like transistors or diodes for proper operation. When the microcontroller sends a signal to the relay module, the relay coil gets energized, closing or opening its internal contacts to control connected devices like lights, motors, or appliances.

3.4 LED

Fig 3.4 LED strip

An LED (Light Emitting Diode) is a semiconductor device that emits light when an electric current passes through it. Unlike traditional incandescent bulbs, LEDs are energy-efficient, durable, and have a long lifespan. They produce light by electroluminescence, where electrons recombine with holes in the semiconductor material, releasing energy in the form of photons (light). LEDs are used in various applications, including display screens, indicators, street lighting, and even in your project for street illumination. They consume less power, generate less heat, and are more environmentally friendly than other lighting sources.

3.5 POWER SUPPLY MODULE

Fig 3.5 Power supply module

A Power Supply Module is a crucial component that provides the necessary electrical power to a system by converting the input power to the required output voltage and current. It typically regulates voltage to ensure stable operation of sensitive components like microcontrollers, sensors, and LEDs. The module can convert AC to DC if needed, using rectifiers, and it often includes features like current limiting and protection against overvoltage, overcurrent, and overheating to safeguard the circuit. In your project, the power supply module ensures that the 8051 microcontroller and other connected devices receive a consistent and safe power source for efficient operation.

CHAPTER 4

RESULTS AND DISCUSSION

OUTPUT

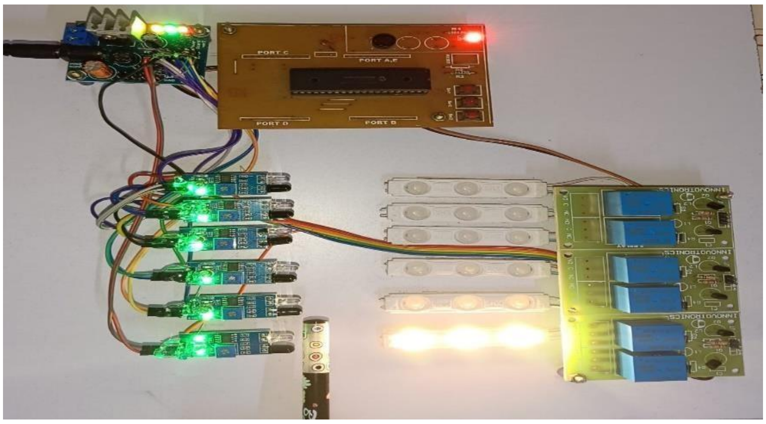

Fig 4.1 First sense

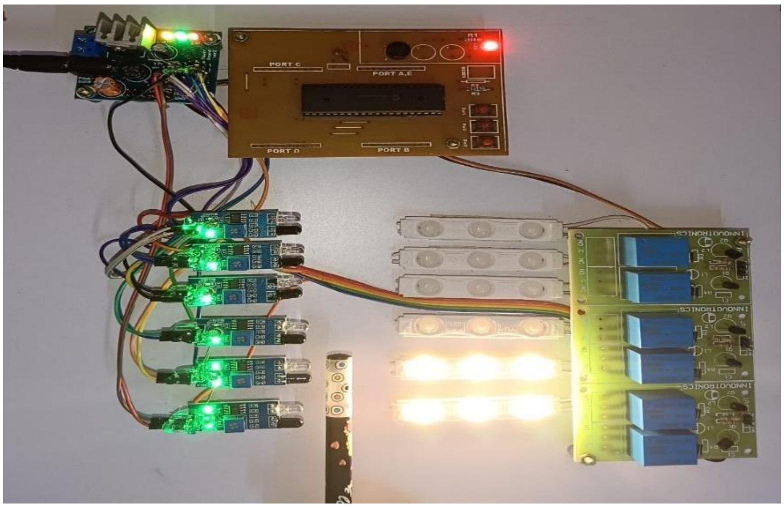

Fig 4.2 Second sense

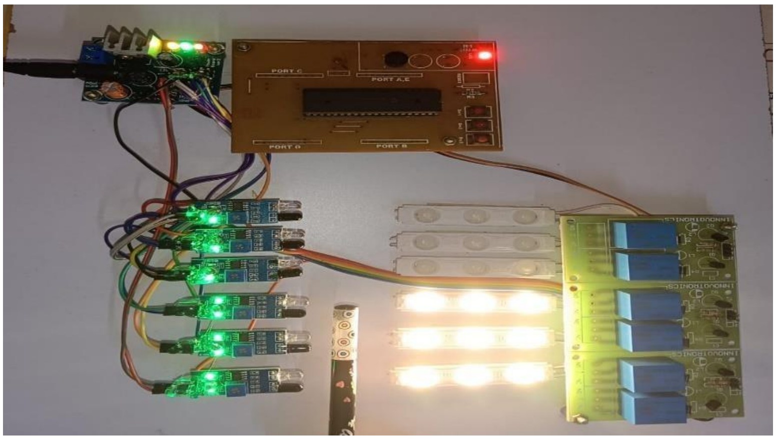

Fig 4.3 Third sense

Automatic Lighting Activation:

- The system successfully activates the corresponding LED (streetlight) when motion is detected by the IR sensors.
- Each LED lights up only in the presence of a nearby object, simulating motion-triggered streetlights.

Energy Efficiency:

- LEDs remain off when no motion is detected, reducing unnecessary power consumption.
- The system demonstrates efficient use of power by lighting only specific sections of the street based on motion.

Quick Response:

- The IR sensors provide a fast response to motion detection, ensuring timely activation of streetlights.

Independent Operation:

- Each sensor and LED pair operates independently, ensuring seamless functionality across multiple sections of the street.

CHAPTER 5

APPLICATIONS

- The system ensures that street lights are only activated when needed, based on vehicle motion, significantly reducing energy consumption compared to traditional street lighting systems that remain on all night.
- The project aligns with the concept of smart cities by integrating automation and real-time response systems to enhance infrastructure efficiency and reduce wastage.
- By optimizing energy use, it can reduce electricity bills for municipalities and local authorities, making it a cost-effective solution for street lighting.
- The system contributes to sustainability by reducing the carbon footprint of street lighting systems, especially if powered by renewable energy sources like solar.
- By activating lights only when vehicles are detected, it improves visibility and safety for drivers and pedestrians, especially in areas with low traffic.
- This system can be scaled for use in various types of environments such as highways, rural roads, parking areas, or residential streets.
- The system can be integrated with Internet of Things (IoT) technologies for remote monitoring and control, allowing authorities to track performance and make adjustments in real time.

CHAPTER 6

CONCLUSION & FUTURE SCOPE

6.1 CONCLUSION

In conclusion, the "Realtime Street Lighting System Activated by Vehicle Motion Using 8051 Microcontroller" is an innovative and practical solution to address energy wastage and inefficiency in traditional street lighting systems. By integrating motion detection, automation, and efficient control through the 8051 microcontroller, the system ensures that street lights are activated only when necessary, significantly reducing power consumption and operational costs. This not only enhances the lifespan of street lighting infrastructure but also contributes to environmental sustainability by minimizing the carbon footprint and light pollution. The project's modular and scalable design allows it to be implemented in diverse real-world applications, such as streets, highways, and parking areas. Overall, this system provides a cost-effective, eco-friendly, and intelligent approach to modernizing street lighting infrastructure, paving the way for smarter, more sustainable urban development.

6.2 FUTURE SCOPE

The "Realtime Street Lighting System Activated by Vehicle Motion Using 8051 Microcontroller" has significant potential for future advancements, making it a scalable and adaptable solution for modern infrastructure. By integrating with smart city frameworks and IoT technology, the system can enable remote monitoring, predictive maintenance, and advanced traffic management. Future developments could include the use of renewable energy sources, such as solar panels, to create a self-sustaining lighting network while further reducing operational costs and environmental impact. Additionally, incorporating advanced sensing technologies like LiDAR or image recognition could improve detection accuracy, accommodating not just vehicles but also pedestrians and cyclists.

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